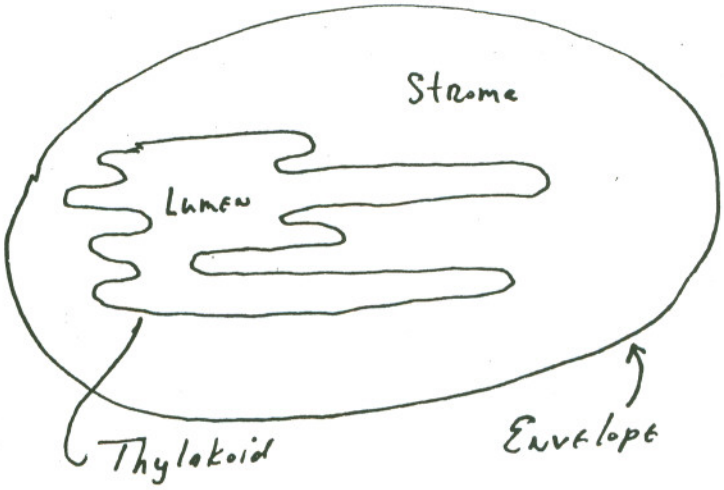
BIOCHEMISTRY CUMULATIVE EXAM

October 3, 1998



A simplified but general view of the chloroplast is shown at the right. The inner and outer membranes comprise what is known as the chloroplast envelope. The photosynthetic apparatus is located on the thylakoid membrane. The aqueous space between the envelope and the thylakoid membrane is referred to as the stroma and the aqueous space inside the thylakoid membrane as the lumen. Although some proteins of the chloroplast are encoded on chloroplast DNA and synthesized within the chloroplast, others are encoded in the nucleus, synthesized on cytoplasmic ribosomes, and then directed post-translationally to the appropriate part of the chloroplast. The genes for these proteins encode, in addition to the information for the mature protein, a signal sequence at the N-terminus which is involved in protein import. To investigate the import of nuclear encoded proteins into the chloroplast, you have cloned the cDNAs for ferredoxin (FD) which is located in the stroma and plastocyanin (PC) which is located in the lumen. In addition, you have constructed hybrid genes in which the signal sequence of the plastocyanin precursor has been replaced by the ferredoxin signal sequence and vice versa. You translate mRNAs from these four genes in vitro, mix the translation products with isolated chloroplasts for a few minutes, reisolate the chloroplasts after proteinase treatment, and then fractionate the chloroplasts to determine into which compartments of the chloroplast the proteins have entered. The products of various steps in the fractionation are examined by SDS electrophoresis followed by electroblotting and immunochemical screening using antibodies against Fd and PC. The results are shown on the next page. The relative shading is representative of the amount of material in each band.

A. (10 points) How efficient is chloroplast uptake of Fd and PC in your in vitro assay? How can you tell?

B. (10 points) Are Fd and PC localized to their appropriate compartments in the chloroplast? How can you tell?

C. (20 points) Are the hybrid proteins imported as you would expect if the N-terminal signal sequences determined their final location? Comment on any unexpected observations and possible explanations for them.

D. (20 points) Why are three bands observed with PCPC and PCFD but only two bands with FDFD and FDPC?

E. (20 points) Based on your experiments, propose a model for the import of proteins into the stroma and lumen compartments of the thylakoid. Be as detailed in your model as your observations allow.

F. (20 points) Cyanobacteria, being prokaryotic organisms, do not have chloroplasts. They do, however, have thylakoid membranes which are ultimately derived from invaginations of the cell membrane. Speculate as to whether signal sequences would be expected to occur in genes for cyanobacterial Fd and PC. Explain your reasoning. If they are not present, indicate why. If they are present, indicate how they might be expected to be similar and/or different from those in the experiment described above.

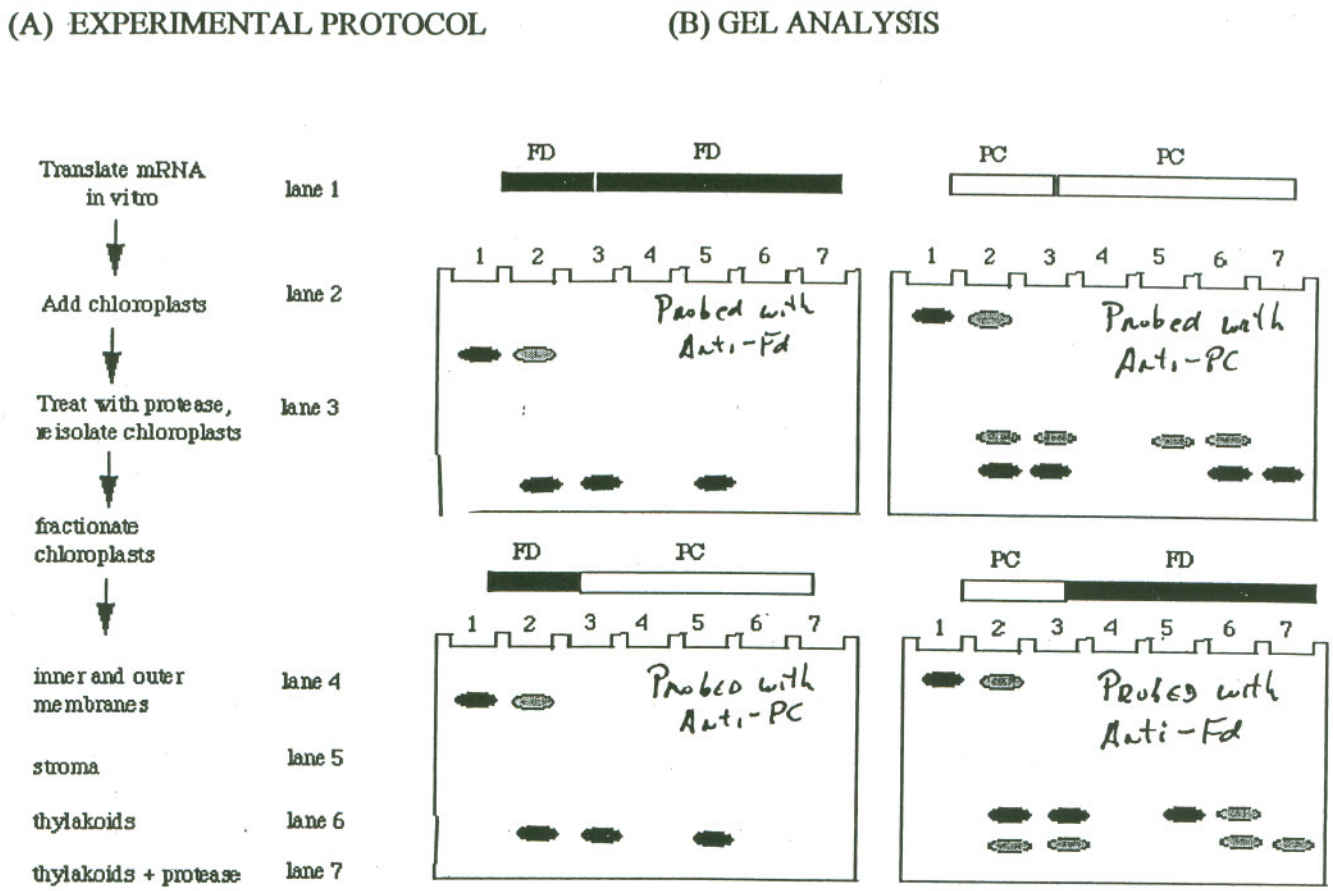


FIGURE HW7 - IMPORT OF FERREDOXIN AND PLASTOCYANIN INTO CHLOROPLAST COMPARTMENTS. Samples were treated as indicated in (A) and then analyzed on gels as shown in (B). Each lane in (B) corresponds to a particular experimental treatment in (A). Ferredoxin gene segments are black and labeled FD; plastocyanin gene segments are white and labeled PC. In all four genes the signal peptide is encoded by the short segment at the left end. The normal ferredoxin gene is FDFD; the normal plastocyanin gene is PCPC.